

1. Introduction to CSS

CSS stands for **Cascading Style Sheets**.

CSS is used to **style and layout HTML elements** on a web page.

HTML defines the structure of a webpage.

CSS controls:

- Colors
- Fonts
- Layout
- Spacing
- Responsiveness

Without CSS, all web pages look plain and unstyled.

2. Why CSS is Required

CSS allows you to:

- Separate content from design
- Reuse styles across pages
- Make websites responsive
- Improve user experience
- Reduce code repetition

CSS follows the principle:

Structure (HTML) + Style (CSS) + Behavior (JavaScript)

3. CSS Syntax

Basic Syntax

```
selector {  
    property: value;
```

```
}
```

Example

```
p {  
    color: blue;  
    font-size: 16px;  
}
```

Explanation:

- `p` is the selector
 - `color` and `font-size` are properties
 - `blue` and `16px` are values
-

4. Ways to Apply CSS

1. Inline CSS

Applied directly to an HTML element.

```
<p style="color: red;">Hello</p>
```

Disadvantages:

- Not reusable
 - Hard to maintain
-

2. Internal CSS

Written inside `<style>` tag in `<head>`.

```
<style>  
  
    p {
```

```
    color: green;

}
```

```
</style>
```

Used for:

- Small projects
 - Single-page styling
-

3. External CSS (Recommended)

Written in a separate `.css` file.

```
<link rel="stylesheet" href="styles.css">
```

```
p {

    color: black;

}
```

Advantages:

- Reusable
 - Clean
 - Professional approach
-

5. CSS Selectors

Selectors define **which elements are styled**.

5.1 Element Selector

```
p {
```

```
    color: blue;
}
```

Applies to all `<p>` elements.

5.2 Class Selector

```
.box {
    background-color: yellow;
}
```

```
<div class="box"></div>
```

- Reusable
 - Starts with `.`
-

5.3 ID Selector

```
#header {
    color: red;
}
```

```
<div id="header"></div>
```

- Unique
 - Starts with `#`
-

5.4 Group Selector

```
h1, h2, p {  
    font-family: Arial;  
}
```

5.5 Universal Selector

```
* {  
    margin: 0;  
    padding: 0;  
}
```

Applies to all elements.

6. CSS Colors

Types of Color Values

```
color: red;  
  
color: #ff0000;  
  
color: rgb(255, 0, 0);  
  
color: rgba(255, 0, 0, 0.5);
```

Used for:

- Text
 - Background
 - Borders
-

7. CSS Background Properties

```
div {  
  
    background-color: lightblue;  
  
    background-image: url("image.jpg");  
  
    background-repeat: no-repeat;  
  
    background-size: cover;  
  
}
```

Common properties:

- background-color
 - background-image
 - background-repeat
 - background-position
 - background-size
-

8. CSS Text Properties

```
p {  
  
    color: black;  
  
    text-align: center;  
  
    text-transform: uppercase;  
  
    line-height: 1.5;  
  
}
```

Text properties include:

- color
- text-align
- text-decoration

- text-transform
 - line-height
 - letter-spacing
-

9. CSS Fonts

```
body {  
  
    font-family: Arial, sans-serif;  
  
    font-size: 16px;  
  
    font-weight: bold;  
  
    font-style: italic;  
  
}
```

Font properties:

- font-family
 - font-size
 - font-weight
 - font-style
-

10. CSS Box Model (Very Important)

Every HTML element is a box.

Box Model Parts

1. Content
2. Padding
3. Border
4. Margin

```
div {  
  width: 200px;  
  padding: 20px;  
  border: 2px solid black;  
  margin: 10px;  
}
```

Understanding the box model is critical for layout design.

11. CSS Border Properties

```
div {  
  border-width: 2px;  
  border-style: solid;  
  border-color: red;  
}
```

Shortcut:

```
border: 2px solid red;
```

12. CSS Margin and Padding

Margin (Outer space)

```
div {  
  margin: 20px;  
}
```

Padding (Inner space)

```
div {  
    padding: 15px;  
}
```

13. CSS Display Property

```
span {  
    display: block;  
}
```

Common values:

- block
 - inline
 - inline-block
 - none
-

14. CSS Positioning

```
div {  
    position: relative;  
}
```

Position types:

- static (default)
- relative
- absolute
- fixed

- sticky

Example:

```
.box {  
    position: absolute;  
    top: 20px;  
    left: 30px;  
}
```

15. CSS Flexbox (Layout System)

```
.container {  
    display: flex;  
    justify-content: center;  
    align-items: center;  
}
```

Flex properties:

- flex-direction
- justify-content
- align-items
- gap

Flexbox is used for one-dimensional layouts.

16. CSS Grid (Advanced Layout)

```
.container {  
    display: grid;
```

```
    grid-template-columns: 1fr 1fr 1fr;

    gap: 10px;
}
```

Grid is used for two-dimensional layouts (rows and columns).

17. CSS Overflow

```
div {

    overflow: auto;
}
```

Values:

- hidden
 - scroll
 - auto
-

18. CSS Opacity and Visibility

```
div {

    opacity: 0.5;
}
```

```
div {

    visibility: hidden;
}
```

19. CSS Pseudo-classes

```
a:hover {  
    color: red;  
}
```

Common pseudo-classes:

- :hover
- :active
- :focus
- :visited

20. CSS Pseudo-elements

```
p::first-letter {  
    font-size: 30px;  
}
```

Examples:

21. CSS Z-Index

```
.box {  
    position: absolute;  
    z-index: 10;  
}
```

Controls stacking order.

22. CSS Media Queries (Responsive Design)

```
@media (max-width: 768px) {  
  
  body {  
  
    background-color: lightgray;  
  
  }  
  
}
```

Used to make websites mobile-friendly.

23. CSS Comments

```
/* This is a CSS comment */
```

24. Common CSS Units

Unit	Meaning
px	Pixels
%	Percentage
em	Relative to parent
rem	Relative to root
vw	Viewport width
vh	Viewport height